FEFF .inp Builder — User Manual

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Quick start

Install Python once; after that it's open, load, check, save. The output is a FEFF6 feff.inp that Artemis opens directly.

1. Install Python 3.8 or newer from python.org. On Windows, tick "Add Python to PATH" in the installer. Nothing else is needed; the window toolkit (tkinter) comes with it.
2. Unzip the download anywhere. On Windows, double-click `feff_inp_builder.pyw`. On macOS or Linux, open a terminal in that folder and run `python3 feff_inp_builder.pyw`.
3. Click **Add files...** and pick a structure file (.cif, .xyz, .pdb, and so on). The `examples` folder has two to try.
4. Check the absorber (it defaults to the heaviest metal) and the summary line at the top right. For `examples/Co_hexaammine.xyz` it reads `nearest: 6 N @ 1.960 A`.
5. Click **Save this one**. The file `<name>_<site>_feff.inp` appears next to the input file.
6. Open that file in Artemis and run FEFF (see Using the file in Artemis).

Optional: run `pip install tkinterdnd2` once in a Command Prompt or terminal to drag files or whole folders onto the window.

Loading structures

A CIF can go in directly; there's no need to open it in Mercury and save it as .xyz first. Symmetry is applied and the molecule is completed automatically.

Format	Extensions	What gets read
CIF	.cif .mmcif .mcif	Cell, symmetry operators and atom sites, expanded to the full molecule or crystal. mmCIF (protein) files use their Cartesian coordinates. Each `data_` block is its own structure.

Format	Extensions	What gets read
SHELX	.res .ins	CELL, LATT, SYMM, SFAC, atoms and PART groups. Q-peaks are ignored.
XYZ	.xyz	First frame. Mercury, Avogadro and ORCA style all work.
PDB	.pdb .ent	ATOM and HETATM lines, first model, altloc A.
Molfile	.mol .sdf .sd	V2000 and V3000. Each SDF record is its own structure.
Mol2	.mol2	The ATOM section. Each molecule is its own structure.
Gaussian	.gjf .com .gau	Cartesian coordinates. Z-matrix input isn't supported.
VASP	POSCAR, CONTCAR, .vasp, .poscar	Direct or Cartesian coordinates.
FEFF	.inp	An existing feff.inp, with its old absorber preselected. Use it to re-center on another atom or re-cut the radius.

- **Several files:** select many in the Add files dialog, or drop a whole folder on the window (needs tkinterdnd2).
- **The list:** each structure gets one row showing its name, format and atom count. **Remove** drops the selected row; **Clear** empties the list.
- **Tags:** atom labels from CIF, SHELX, PDB and mol2 files become the FEFF tags (N5, C18). Formats without labels get Mercury-style ones (C1, C2, N1).
- **Errors:** files that can't be read are listed in a popup with the reason; the rest still load.

Settings

The defaults suit a first-row transition-metal complex: K edge, $S0^2$ 1.0, RMAX 5.0 Å, bonded molecule, no hydrogens. Every change updates the preview immediately.

Setting	Default	What it does
Absorber	Heaviest metal	Element of the absorbing atom (ipot 0).
Site	First site	Which atom of that element. A CIF lists one entry per crystallographic site (Co1, Co2). **All N sites** writes one file per site.
Edge	K	Sets the HOLE card: K = 1, L1 = 2, L2 = 3, L3 = 4.
$S0^2$	1.0	Second number on the HOLE card.
RMAX (Å)	5.0	Longest path half-length FEFF calculates.
Radius (Å)	10	Keeps atoms within this distance of the absorber. 0 means no cut. FEFF6 accepts at most 500 atoms, so use about RMAX + 2 Å for dense solids.
Bonded molecule only	On	Mercury-like: keeps only atoms connected to the absorber through bonds, completed across symmetry and cell edges. A bond is any distance up to the two covalent radii + 0.45 Å.
Everything in radius	Off	Keeps every atom in the sphere, including neighboring molecules, counter-ions and solvent. A radius of 0 is treated as 8 Å here.
Include H	Off	Hydrogens are left out by default; they scatter weakly and mostly add paths.
Drop minor disorder parts	On	Keeps disorder groups 0, 1 and -1 and drops 2 and up. PDB and mmCIF files keep altloc A.
Distance column	Off	Adds each atom's distance after its tag. FEFF ignores the column; it's for reading.

For a molecular complex (a porphyrin, a corrin or another coordination compound), **Bonded molecule only** is the Mercury equivalent. Switch to **Everything in radius** to include lattice neighbors, or when the summary says nothing is bonded to the absorber. Use it too for extended solids such as oxides, and when a long bond (a weak axial ligand, say) is missing from the first shell.

Checking the model

Before saving, check that the summary line's first shell is what the structure should have, for example `nearest: 6 O @ 2.131 A` for examples/CoO_rocksalt.cif.

- **Title line:** the structure name and the absorber site.
- **Summary line:** atom count, number of unique potentials, and the nearest shell grouped by element.
- **Orange notes:** warnings about the model. Each one is explained under Warnings and troubleshooting.
- **Atoms tab:** every atom in the order it will be written: # (its line in ATOMS), tag, element, ipot, distance R and x, y, z. The absorber is highlighted at the top.
- **feff.inp preview tab:** the exact text that will be saved.

Leaving atoms out. Select rows in the Atoms tab (Ctrl-click or Shift-click for several) and click **Exclude / include selected**; double-click or Delete does the same. Excluded rows turn grey, lose their #, and disappear from the feff.inp, with the potentials renumbered. Click the same button again to bring a row back, or **Restore all**.

Use this to make a trimmed model (one quarter of a symmetric macrocycle, for example) or a test model with one ligand removed. Exclusions belong to one structure and site. They last while the window is open and are not saved separately.

Saving

Save this one writes the selected structure. **Convert all** writes every loaded structure, each with its own absorber, site and exclusions, and the settings shown at that moment.

- **File name:** `<structure>_<site>_feff.inp`, for example `complex_Co1_feff.inp`. A CIF with several data blocks adds the block name. If two inputs would give the same name in one run (complex.cif and complex.res), the second gets its extension added (`complex_Co1_res_feff.inp`).
- **Folder:** the **Save to** box. Blank means next to each input file; **...** picks a folder.
- **Overwriting:** if any target file already exists, a prompt asks before anything is written.
- **Contents:** two TITLE lines (structure and absorber; source file and settings), HOLE, CONTROL 1 1 1 1, PRINT 1 0 0 0, RMAX, POTENTIALS, ATOMS and END. The absorber is ipot 0; other elements are numbered by descending Z (N = 1 and C = 2 for Co-N-C models). Atoms are sorted by distance from the absorber.

Using the file in Artemis

The app writes the input file only; FEFF itself runs inside Artemis (Artemis main window guide).

1. In the Artemis main window, click **Add** at the top of the Feff list, or drag the .inp file onto that list (one file at a time).
2. Choose the feff.inp. It opens in the Feff tab of a new Atoms and Feff window.
3. Click **Run Feff** (Feff tab guide).
4. Build the fit from the calculated paths as usual.

- **First check:** the first single-scattering path should be the absorber's nearest neighbor at the distance in the app's summary line.
- **Provenance:** the second TITLE line records the source file, the mode (bonded molecule or all atoms), the radius and whether H was included. It's useful when writing up the methods.

Batch mode (no window)

Add `--batch` to convert from a Command Prompt or terminal without opening the window. Run the commands in the folder that holds the script, or give its full path. Wildcards and folder names work. Use python3 on macOS and Linux; on Windows, py works if python isn't found. Batch mode overwrites existing output files without asking.

```
python feff_inp_builder.pyw --batch examples/Co_hexaammine.xyz
python feff_inp_builder.pyw --batch *.cif -a Co -r 8 -o feff_inputs
python feff_inp_builder.pyw --batch my_folder --site all --crystal
```

Each written file is printed with its atom count and nearest shell, followed by any notes. Without `--batch`, file names on the command line just open the window with those files loaded.

Option	Default	Meaning
`-a`, `--absorber EL`	Heaviest metal	Absorber element
`--site LABEL`	First site	Absorber site label (e.g. Co1); `all` writes one file per site

Option	Default	Meaning
`--edge K|L1|L2|L3`	K	Edge on the HOLE card
`--s02 X`	1.0	$S0^2$
`--rmax X`	5.0	RMAX in Å
`-r`, `--radius X`	10	Keep atoms within X Å; 0 = no cut
`--crystal`	Off	Everything in radius instead of bonded molecule
`--hydrogens`	Off	Keep H atoms
`--keep-disorder`	Off	Keep minor disorder parts
`--distances`	Off	Add the distance column
`-o`, `--outdir DIR`	Next to each input	Output folder

Warnings and troubleshooting

Most notes are informational. The ones that will stop FEFF are overlapping atoms, more than 7 potentials and more than 500 atoms; fix those before saving.

Note or problem	What it means	What to do
Double-click does nothing or opens a text editor	Python isn't installed, or .pyw isn't linked to it	Install Python from python.org, or right-click, Open with, Python
"tkinter is not available"	This Python has no window toolkit	Use the python.org installer (Linux: install python3-tk; Homebrew: python-tk), or use `--batch`
"Nothing is bonded to the absorber"	No atom is within bonding distance in molecule mode (an ionic salt or isolated ion)	Switch to **Everything in radius**

Note or problem	What it means	What to do
"Atoms closer than 0.6 A"	Two positions overlap, usually disorder; FEFF crashes on this	Exclude one atom of each pair listed
"N unique potentials: FEFF6 only allows ipot 1-7"	Too many different elements in the cluster	Lower the radius or exclude distant atoms of rare elements
"Dropped N atom(s) from minor disorder parts"	Disorder group 2+ atoms were removed	Nothing; untick **Drop minor disorder parts** to keep them
"Partial occupancy but no disorder group"	The CIF has half-occupied atoms with no group to choose by	Look for doubled atoms in the Atoms tab and exclude the extras
"Listed a bonded X first..."	FEFF6 builds each potential on the first atom with that ipot; an edge atom with its ligands cut off crashes it, so a bonded one was moved up	Nothing; the file is safe
"Dropped N X atom(s): all beyond RMAX..."	Unbonded edge atoms that can't be in any path were removed	Nothing
"Cluster radius is smaller than RMAX"	Paths longer than the radius are missing	Set the radius to at least RMAX + 2 Å
"N atoms: FEFF6 refuses more than 500"	FEFF6 stops with "Too many atoms in ATOMS list"	Lower the radius below the value the note gives
"No symmetry operators in the CIF"	Only the listed atoms are used, so the molecule may be incomplete	Re-export the CIF from Mercury or the CSD, or export the whole molecule as .xyz
"The bonds repeat through the crystal" or "The bonded fragment is polymeric"	An extended solid (an oxide, a salt, a chain or framework) has no molecule edge, so bonded-molecule mode just fills the radius	Switch to Everything in radius, with a radius of about RMAX + 2 Å